AI Acceptance Across Service Platforms

Text Mining Reddit and Trustpilot: Methods in Practice

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1 The idea and research question

Online services increasingly place AI agents between the user and the thing they came for. Chatbots answer support tickets, algorithms screen and price, automated systems take decisions. We study how people accept or reject that AI, and how acceptance changes with the role AI plays. We compare three service contexts:

- **AI-native services**, where the product itself is an AI agent (ChatGPT, Claude, Replika, Gemini, Perplexity);
- **Peer-to-peer rentals**, where AI is an automation layer on a human marketplace (Turo, Airbnb, Fat Llama);
- **Customer service**, where AI runs the support layer (chatbots replacing agents).

The middle context is the assignment's core: a peer-to-peer sharing economy for high-value professional gear such as cameras and drones (Fat Llama), photography lenses (Lensrentals) and film and event kit (KitSplit), alongside vehicles (Turo, Getaround) and RVs (Outdoorsy). The AI-native and customer-service contexts act as comparison points that show how acceptance shifts as AI moves from being the product, to a marketplace add-on, to the support desk.

> **Research question.** How do users accept and experience AI agents across different service platforms, and what drives or erodes trust as AI moves from being the product, to a marketplace add-on, to the support desk?

Reddit is our discussion baseline, where users describe AI experiences in their own words. Trustpilot is our verification layer, where every review carries a 1 to 5 star rating that acts as a ground-truth acceptance label. Where the open discussion and the labelled reviews agree, we gain confidence. Where they diverge, we learn something.

2 The data

We collected two text sources and used no X or Twitter content.

Reddit (baseline). 4,269 comments from 12 subreddits (November 2023 to May 2026), collected from the Arctic Shift research archive (`01_fetch_reddit_arcticshift.R`) because Reddit's 2026 anti-bot wall blocks live scraping. AI-native subreddits are pulled in full; rental and support subreddits are pulled by AI and automation keywords. Every comment carries a `platform_category`: ai_service 2027, rental 1715, customer_service 527.

Trustpilot (verification). 3,912 reviews from seven peer-to-peer rental platforms (Turo, Fat Llama, RVshare, Outdoorsy, Lensrentals, Getaround, KitSplit), collected with an Apify actor and merged by

`02_merge_trustpilot.R`. Each review has a 1 to 5 star rating, and `06_ai_acceptance.R` flags the ones that describe an AI or automation experience.

3 Method

All analysis runs in one script, `analysis.R`, after data collection. Reddit is mined in depth (steps 1 to 7) and Trustpilot enters as the verification layer (steps 8 to 10). The two classes compared throughout are positive versus negative acceptance: on Reddit inferred from a sentiment lexicon, on Trustpilot read directly from the star rating. We picked methods we could apply meaningfully to this question, and the pipeline covers every technique the brief asks for.

Table 1: The analysis pipeline (one script, analysis.R).

Step	Method	Why
1 Preprocess + classes	Lemmatise + Bing lexicon split	Normalise text; build the two classes
2 Word frequency	Top-10 words per class	What each class is about
3 Word clouds	Commonality + comparison, 1/2-gram	Shared vs distinctive words
4 Sentiment	NRC eight emotions	Emotional texture beyond polarity
5 Co-occurrence network	Bigram network (igraph, ggraph)	Which words travel together
6 Topic models (LDA)	LDA, 4 topics per class, 1/2-gram	Latent themes per class
7 Word embeddings	GloVe word vectors	Word meaning and neighbours
8 Validation	Inferred class vs star rating	Test labels against real stars
9 Themes + role gradient	Shared theme tagging, both sources	Frustrations and the role of AI

We standardised on the **Bing opinion lexicon** (Hu and Liu, 2004) for the positive versus negative split because it ships inside the tidytext package and needs no download, which keeps the pipeline reproducible on any machine. The text mining follows the tidy-text approach of Silge and Robinson (2017).

4 Findings

4.1 What each class talks about (word frequency and clouds)

After normalising the text (lowercasing, removing punctuation, numbers and stop words, then lemmatising to base words), we count the most frequent words in each class. Frequency is the fastest read on what each class is about. The negative class centres on *airbnb, host, guest, bot, contact, issue*; the positive class on *host, guest, automate, message, support* (Figure 1). The word *automate* is a top positive term: hosts praise the automation that saves them time, the same automation that, when it fails, drives the negative class. The comparison cloud (Figure 2) shows the words that most separate the two classes.

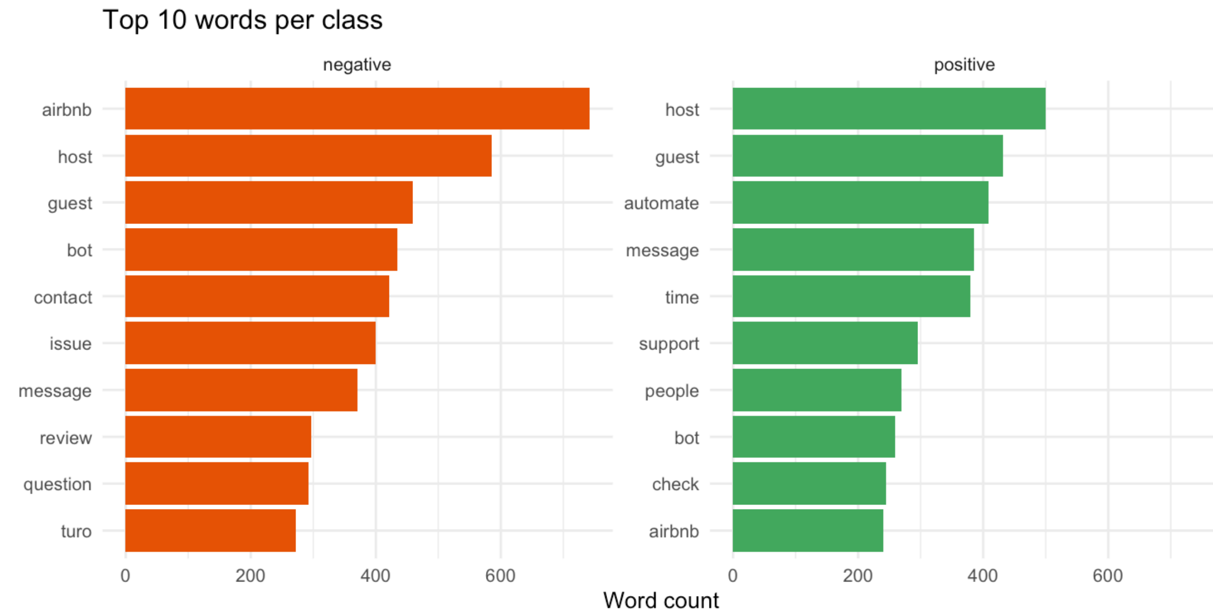

Figure 1: Top 10 words per class.

Figure 2: Comparison cloud: words coloured by the class they belong to.

4.2 How the classes feel (sentiment)

The two classes are split by polarity, so to add information we score the **NRC emotion lexicon** (Mohammad and Turney, 2013), which tags words with eight emotions rather than a single positive or negative value. The positive class scores higher on trust, anticipation and joy; the negative class higher on sadness, fear, anger and disgust (Figure 3). The gap is widest on **trust**, which is the currency of a sharing economy.

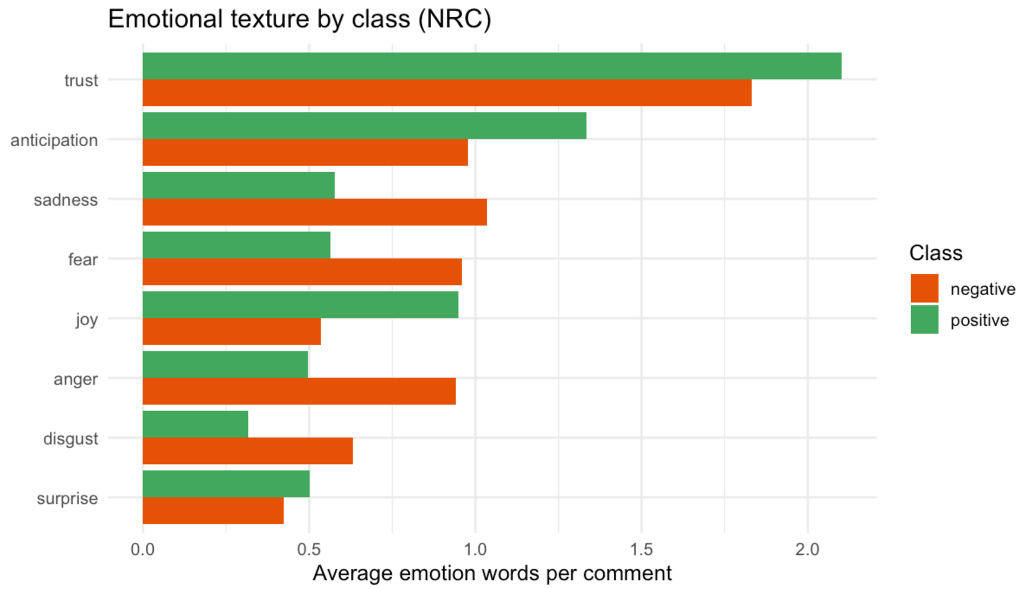

Figure 3: Average emotion words per comment, by class (NRC lexicon).

4.3 How words connect (co-occurrence network)

Frequency lists lose structure, so we count adjacent word pairs (bigrams) and draw the strongest as a network with igraph and ggraph. A network shows which words travel together. A clear core runs from *email* to *customer, service, support* and *contact*, and a chain runs *agent, human, live, person, real*: users asking for a real person (Figure 4). A second cluster, *automate, message, guest, host*, captures host-side automation.

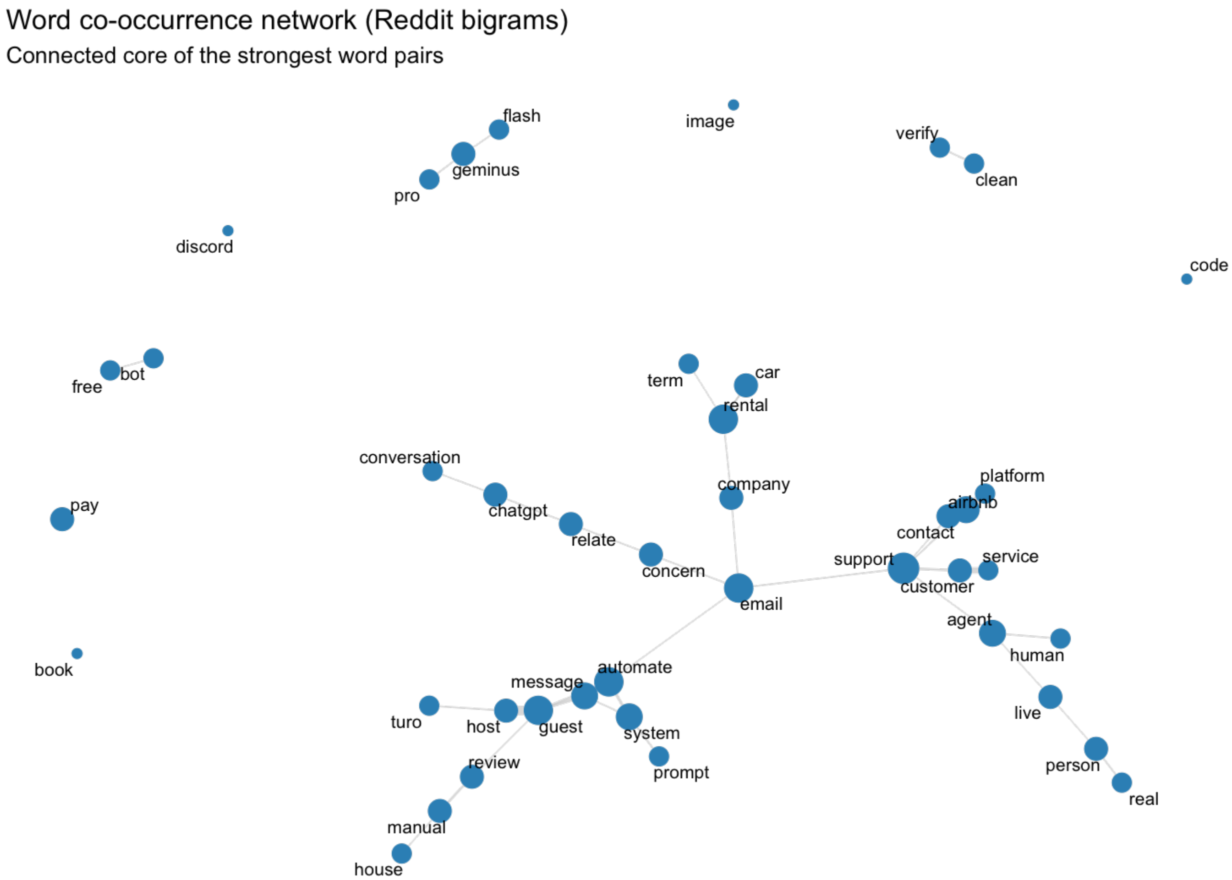

Figure 4: Word co-occurrence network of the strongest Reddit bigrams.

4.4 The latent topics (LDA)

Latent Dirichlet Allocation (Blei, Ng and Jordan, 2003) groups co-occurring words into topics. We fit four topics per class on unigrams and bigrams, which summarises thousands of comments into a few readable subjects. For the negative class (Figure 5), Topic 1 is contacting support and bots on Airbnb, Topic 2 is Turo car rentals, Topic 3 is hosting, reviews and automated booking, and Topic 4 is human versus bot answers. Support contact and automation recur across topics.

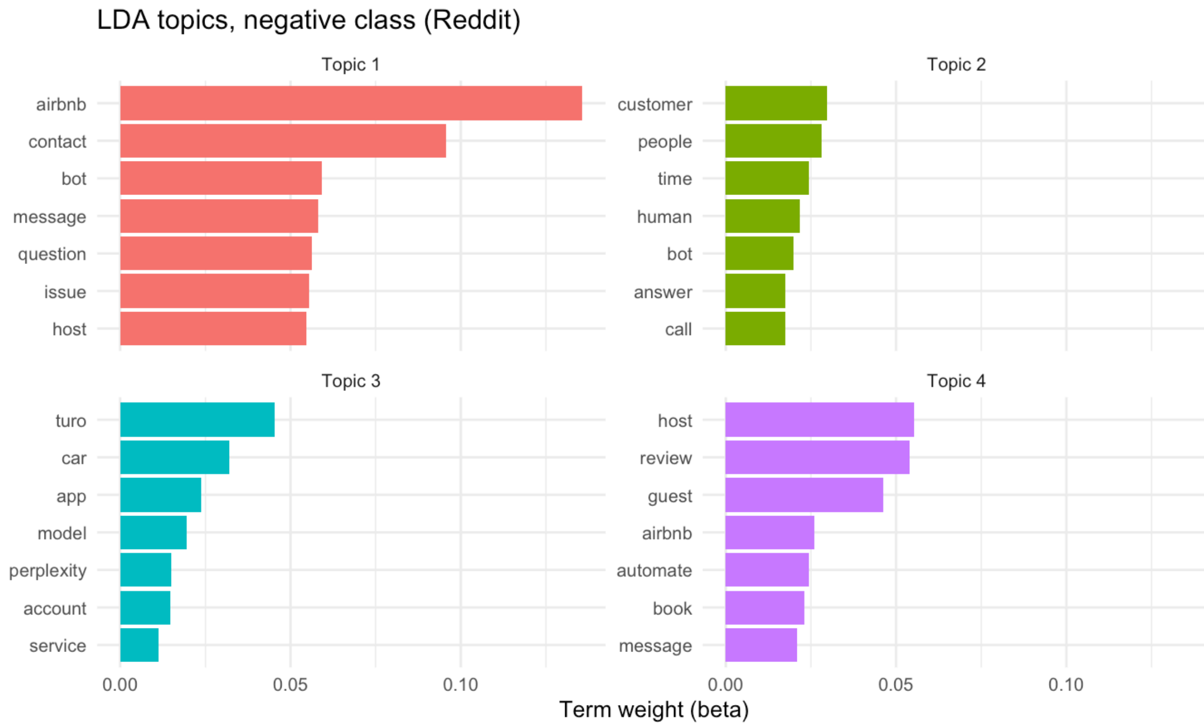

Figure 5: LDA topics for the negative class (top terms per topic).

4.5 The meaning behind words (embeddings)

Lexicon and topic methods ignore meaning. **GloVe** word embeddings (Pennington, Socher and Manning, 2014) place words in a vector space learned from the corpus, so we can read the nearest neighbours of key terms. The vectors independently rediscover the human-versus-automation tension at the heart of the project:

- **trust**: safety, people, stop, relationship, customer, design, team, build
- **bot**: contact, question, phone, support, concern, issue, customer, chat
- **human**: real, agent, customer, replace, automation, actual, speak, support
- **host**: guest, airbnb, platform, communication, experience, turo, property, automate
- **automate**: message, guest, set, check, system, time, send, book

The neighbours of *human* (agent, real, loop, support, automation) capture users asking for a real person, while *bot* sits among support and contact channels.

4.6 Does the inferred class hold up? (validation)

Reddit has no ground-truth label, so the positive or negative class is *inferred* from sentiment. Trustpilot has both text and a star rating, so it is where we can test that inference: we apply the same lexicon class to Trustpilot and compare it to the star-based class. Agreement is **93.4%** (Cohen's $\kappa = 0.76$, n = 3,534), which is substantial by the usual benchmark, and the lexicon recovers 76% of genuinely negative reviews (Figure 6). This is the bridge that lets us read the two corpora together with confidence.

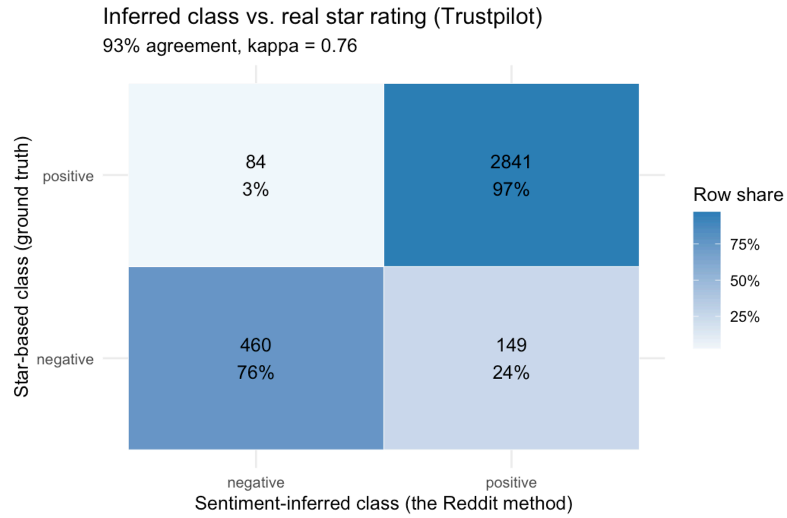

Figure 6: Inferred class vs. the real star rating on Trustpilot.

4.7 The big pattern (the role of AI)

With the method validated, we tag both corpora with the same six themes and measure acceptance by AI's role. On Reddit, the positive share is highest where users chose the AI product and lowest where AI is a rental add-on:

Table 2: Reddit: positive-class share by the role of AI.

Context	Comments	Positive share
AI-native (the product)	1279	58%
Customer service (support desk)	401	56%
Rental (a marketplace add-on)	1321	49%

Trustpilot makes the same point with hard numbers: rental reviews that mention AI or automation average **2.32 stars** against **4.21** for the rest, a **1.89-star** gap (Figure 7). Same technology, opposite reception, depending on whether the user chose it. Figure 8 shows which themes drive the negative class, and that four of them do so in both sources at once.

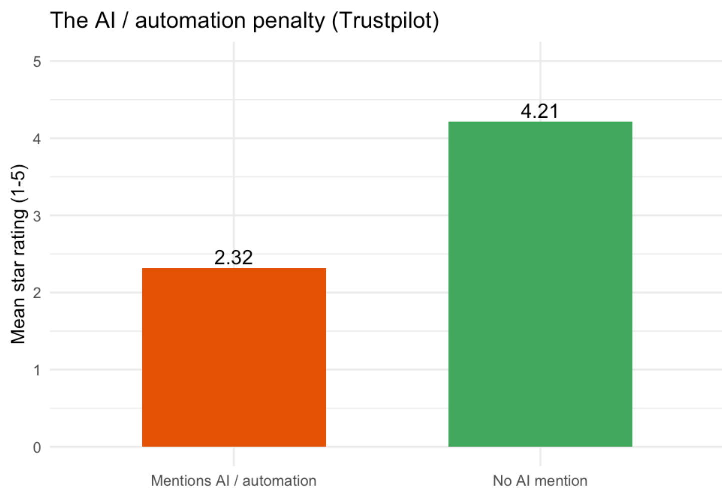

Figure 7: The AI and automation penalty on Trustpilot.

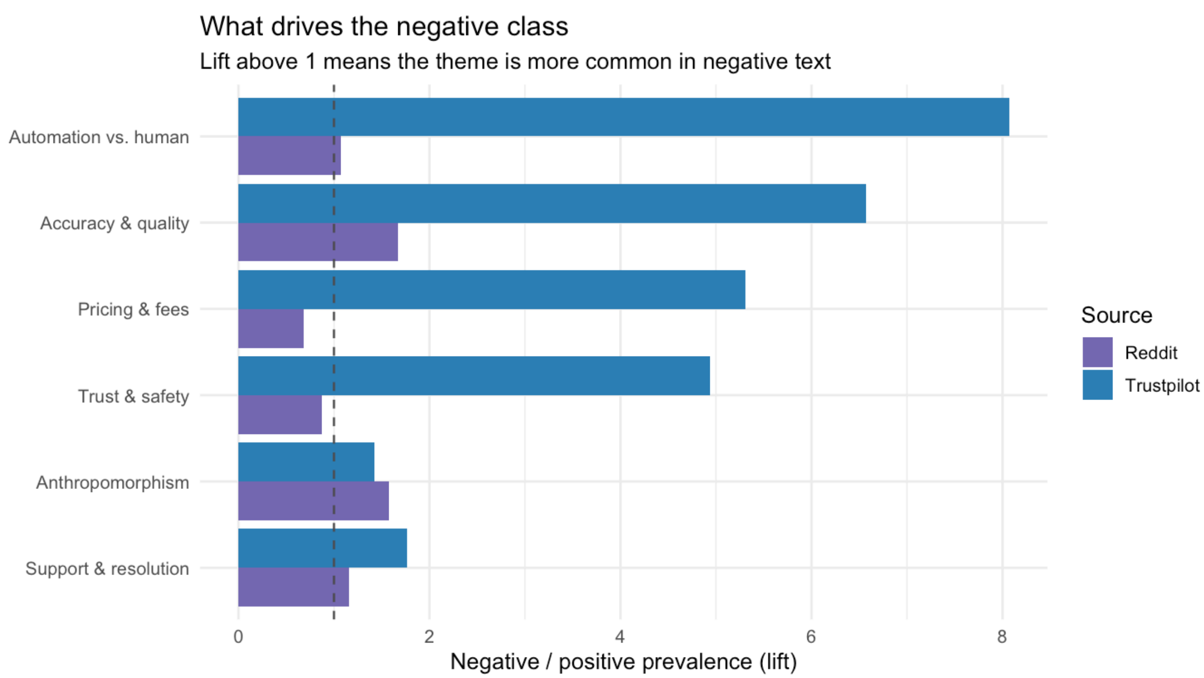

Figure 8: Themes that drive the negative class. Lift above 1 means more common in negative text.

5 Key differences between the two classes

Across every method, the two classes separate along one axis: **control**.

Positive class	Negative class
Praises automation that saves time (automated messages, quick checks)	Hits forced automation and dead ends (*bot, contact, issue*)
Trust, joy, anticipation	Anger, fear, sadness, disgust
Talks about value and convenience	Repeatedly asks for a human (*agent, real, person*)

Positive class	Negative class
AI was **chosen**	AI was **imposed**

Acceptance is high when the user chose the automation and can still reach a human. It collapses when automation is imposed and the human exit is missing. This is the practical meaning of *algorithm aversion* (Dietvorst, Simmons and Massey, 2015): people tolerate AI for tasks they see as mechanical and resist it for tasks they see as needing human judgement.

6 Strategic recommendations

The evidence converts into a short, ordered list of design rules. They are written for a product or marketing team deciding how an AI agent should behave.

1. **Keep a human one click away.** Automation / 'no human' is the strongest marker of negative reviews (8x on Trustpilot) and the top negative topic on Reddit.
2. **Disclose the AI and admit its limits.** Accuracy and 'doesn't work' complaints separate the classes most cleanly (negative class 1.8x on Reddit, 6.9x on Trustpilot).
3. **Optimise for first-contact resolution.** Support and resolution language drives the negative class in both sources (1.2x Reddit, 1.8x Trustpilot).
4. **Match the persona to the role.** Anthropomorphism is welcomed when AI IS the product but reads as cold in transactions; Reddit is most positive for AI-native, least for rentals.
5. **Show all-in price and refund rules up front.** Pricing and fees dominate negative rental reviews (5.4x on Trustpilot); a rental-specific, not AI-specific, frustration.

The first rule matters most: across both data sources, the strongest single signal is that being trapped with a bot, unable to reach a person, destroys acceptance, while automation the user opted into earns it. An AI agent should be **transparent** (labelled as AI, honest about limits), **escapable** (a human is one click away) and **role-matched** (personable when it is the product, efficient and unobtrusive when it is a transactional add-on).

7 Methods, tools and limitations

The analysis is written in R. Preprocessing uses tidytext and textstem; sentiment uses the Bing and NRC lexicons (via tidytext and syuzhet); the network uses igraph and ggraph; topic models use topicmodels; embeddings use text2vec. The report is fully reproducible: `analysis.R` regenerates every figure and table that appears here.

A few limits are worth stating plainly:

- Trustpilot reviews skew positive, which is typical of opt-in review sites.
- The Bing lexicon is binary, so the inferred class is a coarse instrument; the 93.4% agreement with stars shows it is good enough for classification, but exact magnitudes should be read as directional.
- The Trustpilot side covers rentals only, so the AI-native end of the role gradient rests on Reddit. The rental contrast, the core of the sharing economy, is the part validated across both sources.
- Trust appears mostly implicitly (bot or human, automated, algorithm), so we read it through themes and emotion rather than the literal word. English only.

8 References

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